

Question Paper

Exam Date & Time: 26-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: ICT 3122/IT_3122 PRINCIPLES OF OPERATING SYSTEMS
(SPL: INFORMATION TECHNOLOGY)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A)

Consider a warehouse robot system where multiple robots update inventory status in a database. Apply the concept of critical section to ensure accurate stock levels in the inventory database.

Course Outcome: "CO2" / Cognitive Level: "Application"

(5)
- 1B)

A user program attempts to read data from a file using the read() system call. The CPU first executes the user program, then switches to a privileged mode to carry out the file access operation, and finally returns control back to the user program after completion. Identify and explain the operating system mechanism that enables this secure transition between user-level and kernel-level execution.

Course Outcome: "CO1" / Cognitive Level: "Application"

(3)
- 1C)

Analyze how the four necessary conditions for deadlock interact to create deadlock situation. Illustrate your explanation with a neat, diagram showing the relationship between processes and resources.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

(2)
- 2A)

Justify the use of a counting semaphore instead of multiple binary semaphores in a producer-consumer system with a large buffer. Support your answer with suitable pseudocode.

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

(5)
- 2B)

A web server needs to handle multiple client requests simultaneously. Demonstrate how the use of fork() and exec() system calls can be applied to achieve efficient multitasking in this situation.

Course Outcome: "CO1" / Cognitive Level: "Application"

(3)
- 2C)

A UNIX-based server is running 6 processes with the following memory and CPU characteristics as shown in Table Q. 2C.
The system has a total of 1000 MB of physical memory. Currently, 920 MB (92% of total) is in use, leading to high memory exhaustion. As a result, the system has started experiencing thrashing, the medium-term scheduler is tasked with swapping out processes to reduce memory load.

(2)

Table Q. 2C

Process	Memory (MB)	CPU (%)	I/O (MB/s)	Priority
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Process	Memory Usage (MB)	CPU Usage (%)	I/O/CPU Bound	Priority
P0	120	10	I/O Bound	2
P1	250	80	CPU Bound	5
P2	150	20	I/O Bound	3
P3	300	70	CPU Bound	4
P4	200	15	I/O Bound	1
P5	100	50	CPU Bound	2

(a) Identify which process(es) should be swapped out to bring memory usage below 80%. Justify your selection using CPU vs I/O usage and priority (higher number = higher priority).

(b) Calculate the total memory freed if your selected processes are swapped out.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) A system detects a deadlock involving five processes shown in Table Q. 3A. The OS collected the following facts about each process (% of progress of execution, priority, resources held, and resources requested). Analyze and propose a recovery action with justification: (Priority values are assigned such that the largest number indicates the highest priority, and the smallest number indicates the lowest priority). (5)

Table Q. 3A

Process	% Completed	Priority	Resources Held	Resources Still Needed	Type
P1	85%	5	4	1	Interactive
P2	70%	4	3	2	Interactive
P3	25%	2	2	3	Batch
P4	20%	1	3	2	Batch
P5	30%	1	1	1	Interactive

(a) Apply an appropriate victim-selection rule that aims to reduce the loss of completed work and minimize disruption to user processes. Identify the process that should be selected as the victim and justify your choice.

(b) For the long-running interactive processes P1 and P2, determine whether process termination or resource preemption provides a more effective deadlock recovery strategy that ensures minimal cost or guaranteed resolution. Justify your choice.

(c) Suggest one concrete step the OS can take after recovery to maintain consistency.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3B) Suppose that the processes P1, P2 and P3 arrive for execution at the times indicated in Table Q. 3B. (3)

Table Q. 3B

Process	Arrival Time	Burst Time
P ₁	0	8
P ₂	0	4
P ₃	1	1

(a) Compute the average turnaround time average waiting time for these processes with the non-preemptive SJF scheduling algorithm?

(b) Calculate the number of context switches.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3C) A system has $m = 300$ physical frames. The operating system reserves 12 frames, and the remaining frames are allocated to 4 user processes with the following page requirements: $s_1=120$, $s_2=60$, $s_3=200$, $s_4=100$ (2)
- a) Compute the number of frames allocated to each process using proportional allocation.
- b) Compute the number of frames allocated to each process using equal allocation. Discuss how equal allocation may lead to underutilization of frames by smaller processes and increased page fault rates for larger processes.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) A developer is building a large software system that contains multiple routines, but only a few are used during typical execution. (5)
- (a) Evaluate whether dynamic loading and dynamic linking would be more beneficial than static linking for this project.
- (b) Explain how the use of a stub in dynamic linking helps optimize program size and runtime behavior.
- (c) Identify which component of the system is responsible for bringing an unused routine into memory when it is called at runtime.

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- 4B) A media production server manages large video files on a disk with 500 cylinders. The server is currently using the First-Come, First-Served (FCFS) disk scheduling algorithm. The disk head is currently located at cylinder 250. (3)
- The queue of pending requests is: 480, 50, 400, 100, 320, 110, 200, 350
- (a) Calculate the total head movement (in cylinders) required to service all requests using the currently implemented FCFS disk scheduling algorithm.
- (b) Using the same queue, calculate the total head movement (in cylinders) if the server uses the SCAN disk scheduling algorithm, assuming the head initially moves toward cylinder 499. Based on your calculations, compare the two algorithms and state which one is the most efficient for this request queue.
- (c) Does starvation occur in FCFS and SCAN disk scheduling algorithms? Justify your answer.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4C) A university's file system allows both faculty and students to access a single shared project document without keeping duplicate copies in their own directories. Identify which directory structure that best suited for this situation. Justify your answer. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) Consider the deadlock situation that can occur in the dining philosophers problem when the philosophers obtain the chopsticks one at a time. Discuss how the four necessary conditions for deadlock hold in this setting. Discuss how deadlocks could be avoided by eliminating any one of the four necessary conditions. Demonstrate the use of wait() and signal() operations inside a monitor for process coordination. (5)

Course Outcome: "CO3" / Cognitive Level: "Create"

- 5B) Consider the following set of periodic tasks with their execution times, deadlines, and periods as shown in Table Q.5B: (3)

Table Q.5B

Tasks	Execution Time	Deadline	Period
T_1	3	5	10

T1	2	3	10
T2	2	6	12
T3	2	4	6
T4	1	3	4

- (a) Calculate the hyperperiod.
(b) Check whether the task set is schedulable under EDF.
(c) Show the Gantt chart if the tasks can be scheduled using EDF scheduling algorithm.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) A team is building OSes for two devices: (2)
Device C: An implantable cardiac pacemaker that triggers heart pulses.
Device D: A live sports streaming app on smartphones that adapts video quality. Suggest the most suitable type of operating system required for Device C and D. Justify your choices with appropriate reasons.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_3122:PRINCIPLES OF OPERATING SYSTEM

#	Course Outcome	Description	Marks
1	1	CO 1	6
2	2	CO 2	27
3	3	CO 3	12
4	4	CO 4	5
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_3122:PRINCIPLES OF OPERATING SYSTEM

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	31(62%)
4	Analysis	2	4(8%)
5	Create	2	10(20%)
6	Evaluate	1	5(10%)
Total		15	50 (100%)